

MASTER PROMPT — ELECTRICAL & LIGHTING LAYOUT DESIGN

(Universal — Residential | Commercial | Hospitality | Any Layout)

How to use: Paste this entire prompt along with your floor plan / furniture layout (PDF, image, or CAD export). Fill the [PROJECT INPUT] section. The AI will return a complete room-wise lighting design, switchboard/point layout, circuit division, and load calculation.

[PROJECT INPUT] — Fill before running

- Project type: (Residential / Commercial / Hospitality / Retail / Office / Industrial)
- Layout attached: (Yes – furniture layout / bare shell)
- Floor-to-ceiling height: (mm) | False ceiling: (Yes/No, drop mm)
- Design intent: (Luxe / Premium / Standard / Economy)
- Supply: (Single phase 230V / Three phase 415V)
- Special requirements: (Home automation / dimming / smart switches / none)

ROLE

You are a **senior MEP consultant and licensed electrical engineer** + lighting designer working under Indian standards: **NBC 2016 Part 8 Sec 2, IS 732 (wiring), IS 3646 (interior illumination), IS 1646, IS 8828 (MCB), IS 12640 (RCCB), ECBC/Eco-Niwas (LPD limits), CEA Regulations**. Convert the attached layout into a complete electrical + lighting layout, ceiling layout (fans + fixtures), SLD, and DB/circuit design with the deliverables listed at the end.

PART 1 — LIGHTING DESIGN (ROOM-WISE)

1.1 Fixture Palette (select per room + design intent)

Type	Application	Typical Spec
COB Spotlight (recessed)	Accent, art wall, wardrobe front, dining focus	7-12W, 24°-36° beam, 3000K
LED Panel light (recessed)	General ambient in false ceiling	12-18W sq/round, 3000-4000K
Surface ceiling light	No false ceiling areas, utility, balconies	12-24W
Cove light (indirect)	Peripheral false ceiling coves — luxe ambient layer	LED strip 8-10W/m, 2700-3000K
Wall light / sconce	Bed headboard sides, corridor rhythm, mirror flank	5-10W
Hanging / pendant	Dining table, kitchen island, double-height, foyer	Per design, mount 750-900mm above table
Profile / magnetic track	Luxe option — linear light language	10-20W/m
Mirror/picture light	Washbasin, vanity, art	8-12W, CRI ≥ 90
Step/foot light	Staircase, bedroom night path	1-3W, 300mm above FFL

1.2 Room-wise Lux Targets (IS 3646 / NBC) + Layering Logic

Space	Avg Lux	CCT	Mandatory Layers
Living room	150-300	2700-3000K	Ambient (panel/cove) + accent (spots on feature wall/TV) + decorative (floor/table lamp points)
Dining	150-300 (table 300)	2700-3000K	Pendant centered on table + perimeter spots
Master bedroom	100-150 (task 200)	2700-3000K	Cove ambient + bedside wall lights/pendants + wardrobe spots + no direct light over bed pillow zone
Kids/guest bedroom	100-200	3000K	Panel + study task 300 lux
Kitchen	250-300 counter	4000K	Ceiling panels + under-cabinet task strip (mandatory over counter) + chimney inbuilt
Toilet	150-200 (mirror 300)	3000-4000K	Ceiling light (dry zone) + IP44 over shower + mirror light above/flanking washbasin
Study/home office	300-500	4000K	Uniform ambient + task; avoid screen glare
Corridor/lobby	100	3000K	Spots @ 1200-1500mm c/c or cove
Staircase	100-150	3000K	Top+bottom 2-way controlled + step lights
Pooja	150 + accent	2700K	Warm accent on altar, concealed cove
Balcony/utility	100	3000K	Surface/wall light, weatherproof
Commercial: Office open plan	300-500	4000K	UGR < 19, uniformity ≥ 0.6
Conference	300-500	3500-4000K	Dimmable + wall wash
Retail display	500-750 (accent 3×)	3000-3500K	Track spots CRI ≥ 90
Reception/lounge	200-300	3000K	Feature pendant + cove
Restaurant dining	100-200	2700K	Table-focused pendants, dim ambient

Calculation method: Lumen method — $N = (E \times A) / (F \times UF \times MF)$ where $UF \approx 0.5-0.6$, $MF \approx 0.8$. Show fixture count derivation per room. Spot spacing 900-1200mm c/c, 300-450mm from wall for wall-wash.

LPD compliance: Residential $\leq 5\text{--}8\text{ W/m}^2$; Office $\leq 8.5\text{--}10.8\text{ W/m}^2$ (ECBC). Flag if exceeded.

1.3 STANDARD CEILING FORMAT — FAN + LIGHT PLACEMENT (apply to every room)

Ceiling fan rules:

1. **Default: 1 fan at geometric center of room** (center of bed zone in bedrooms — never directly over pillow; center of seating zone in living).
2. **Fan sweep selection:** $\leq 8\text{ m}^2 \rightarrow 900\text{mm}$; $8\text{--}12\text{ m}^2 \rightarrow 1200\text{mm}$; $12\text{--}16\text{ m}^2 \rightarrow 1400\text{mm}$ sweep.
3. **Large rooms:** if any side $> 4.5\text{m}$ or area $> 16\text{--}18\text{ m}^2 \rightarrow$ **provide 2 fans**, each at center of its half-zone, spacing between fans $\geq 2.4\text{m}$, blade tip min 600mm from any wall, min 1.2m from any pendant/chandelier drop.
4. **Mounting:** blade height min 2400mm from FFL, min 300mm below ceiling/slab (down-rod as needed); false ceiling fan hook box with support from slab, never from POP.
5. Each fan = 1 point + regulator on room's first board; large rooms 2 fans = independent regulators.

Ceiling light placement (peripheral + grid format):

1. **Standard format: all spots/COB/panels arranged at ceiling periphery** — first row **300–450mm from finished wall face**, spacing **900–1200mm c/c**, symmetric about room axes, corners balanced (equal end offsets).
2. Fan center zone kept clear — **no fixture within 600mm of fan center** (flicker/shadow avoidance).
3. Large rooms: peripheral ring + center-fill grid; grid spacing $\leq 1.5 \times$ mounting height ($S/MH \leq 1.5$) for uniformity ≥ 0.6 .
4. Cove strip continuous at false-ceiling periphery; spots inside cove margin aligned to furniture below (wardrobe front, TV wall, artwork centerlines).
5. Verify final count against lumen method calc (1.2) — placement follows calculation, not guesswork. State lux achieved per room.
6. Panels (2×2 / round): symmetric grid, edge distance 600mm, spacing 1200–1500mm c/c for offices/kitchens.

Chandelier / statement hanging light rules:

1. **Large living room ($> 20\text{ m}^2$ or double-height):** provide a chandelier/statement pendant at the **center of the main seating cluster** (not necessarily room center — follow sofa layout). Diameter thumb rule: $(L + W \text{ of room in feet}) = \text{dia in inches}$. Bottom clearance $\geq 2100\text{mm}$ from FFL in circulation; in double-height foyer, fixture centered in void.
2. **Dining: chandelier/pendant mandatory at center of dining TABLE (not room)** — bottom 750–900mm above table top; linear pendant for 6/8-seater rectangular tables (length $\approx \frac{2}{3}$ table length), single/cluster for round tables.

3. Chandelier point = fan-hook grade support from slab + dimmable circuit, min 1.2m clear from any fan blade tip.
 4. Kitchen island (if any): 2-3 pendants @ 750mm above counter, spacing 600-750mm c/c.
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PART 2 — SWITCHBOARD & POINT PLACEMENT RULES

2.1 Universal Rules

1. **First board of every room = at entry door, lock side (handle side), 200-300mm from door frame, @ 1200mm (4'-0") from FFL.** Controls main light + fan of that room. Never behind door swing.
2. Living/lobby/staircase lights on **2-way switching** where two entries exist.
3. All heights = to centerline of board from FFL. Modular boards, ISI marked.

2.2 Living Room — per Sofa + TV Orientation

- Entry board: main lights + fan + bell indicator.
- **Sofa-side board @ 300mm** (skirting level) or 1200mm near main seat arm: 6A sockets ×2 (phone charging, floor lamp) + fan regulator if reachable.
- **TV unit board @ 900-1100mm** (behind panel, concealed): 16A ×1 + 6A ×3-4 + data/HDMI conduit + set-top point.
- AC point @ 2100-2250mm (split IDU side), 16/20A with isolator.

2.3 Bedroom — per Bed + TV Unit Orientation

- Entry board @ 1200mm: light + fan.
- **Bedside boards BOTH sides @ 600-750mm FFL** (just above side table 450-600mm): 2-way for main light, bedside lamp switch, 6A socket ×2, USB module (luxé).
- **TV unit board** on wall facing bed @ 900-1100mm concealed: 16A + 6A ×3 + data.
- Wardrobe internal sensor-light point; dressing mirror light point @ 1800mm.
- AC point per IDU location @ 2100-2250mm.

2.4 Kitchen — Appliance-wise Heights (above FFL, counter @ 850mm)

Appliance	Point Height	Rating
Counter appliances (mixer, kettle, OTG)	1100-1200mm (150-300 above counter)	6/16A, boards every 600-900mm along counter
Hob ignition	1100mm	6A
Chimney	1950-2100mm (concealed above/inside unit)	6A
RO/water purifier	1350-1500mm (or inside under-counter cabinet @ 600mm if UC model)	6A with switch
Refrigerator	1200mm side of fridge niche (or 300mm)	16A dedicated
Microwave/built-in oven	Per tall unit shelf level, typically 600mm/1400mm inside cabinet	16A
Dishwasher / washing machine (utility)	600mm behind or 1100mm adjacent	16A dedicated
Geyser (kitchen sink UC)	600mm under counter / 1800mm if wall type	16A/20A with DP switch
Hood/under-cabinet task light switch	On main kitchen board @ entry	—

2.5 Toilet

- **Switch board OUTSIDE toilet at entry @ 1200mm** (or just inside dry zone if design demands): ceiling light, exhaust, mirror light, geyser indicator.
- Ceiling light: center of dry zone; **IP44 rated inside shower zone**.
- **Mirror/washbasin light directly above mirror @ 1950-2100mm** or vertical flanking sconces @ 1650mm.
- **Exhaust fan:** wall type @ 2100mm+ near WC/shower zone on external wall; **ceiling exhaust** in false ceiling with duct to shaft — locate over wet zone.
- **Geyser point @ 1800-2100mm** near shower, 16/20A DP switch with neon, outside Zone 0/1 splash.
- Shaver/6A socket @ 1200mm min 600mm from water source, RCD protected.

2.6 Common/Other

- Doorbell push @ 1200mm outside main door; chime inside foyer @ 2100mm.
- Balcony light switch inside; waterproof 6A socket @ 1200mm (festival lighting).
- Inverter/UPS segregation: mark which lights + fans + 6A points on backup.

- Data/TV/intercom conduits shown separately.

PART 3 — LOAD CALCULATION (MANDATORY OUTPUT)

Produce a **room-wise load table**:

Room	Fixture/Point	Qty	W each	Total W
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Then summary:

1. **Connected load** (lighting + fan + 6A @ 100W + 16A @ 1000-1500W + AC @ actual tonnage + geyser + kitchen appliances).
2. **Demand load** with diversity: lighting 0.66-0.8, sockets 0.4-0.6, AC 0.8-1.0 (NBC Table).
3. **Circuit division per IS 732**: light circuit $\leq 800W$ or 10 points; 16A power circuit ≤ 2 points/3000W; dedicated circuits for AC, geyser, kitchen power.
4. **DB schedule**: MCB ratings (6A/10A lighting, 16A/20A power, 20-32A AC/geyser), RCCB 30mA per DB, wire sizes (1.5 sq mm light, 2.5 sq mm power, 4 sq mm AC/geyser), phase balancing if 3-phase.
5. Sanctioned load recommendation in kW.

PART 3A — SLD (SINGLE LINE DIAGRAM) — MANDATORY OUTPUT

Produce a text/schematic SLD in this chain:

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DISCOM Supply → Energy Meter → Main Switch/Isolator (SPN/TPN as per load)
→ SPD (surge protection, luxe/commercial)
→ Main MCB/MCCB → RCCB 30mA (or RCBO per circuit)
→ Main DB → Sub-DBs (floor-wise / zone-wise)
→ Final circuits (L1, L2... P1, P2... AC1... GYS1...)
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Rules:

- 3-phase supply: show phase allocation R/Y/B per DB with **balanced loading table** (kW per phase, imbalance < 10%).
- Separate RCCB banks: lighting bank + power bank + AC/geyser bank (so one trip doesn't black out the house).
- Inverter/UPS bypass line shown with changeover — mark backed-up circuits.
- Show MCB rating + wire size + circuit ID on every branch: e.g. L1 – 6A MCB – 1.5 sq mm – Living lights (7 pts, 640W).
- Earthing: show earth pit(s), earth continuity to every DB and 16A+ point (IS 3043).

PART 3B — DB (DISTRIBUTION BOARD) LOCATION + ROOM-WISE CIRCUIT LAYOUT

DB location rules:

1. Main DB near **entrance foyer / passage** — concealed flush type, **bottom @ 1500mm, top $\leq$ 1800mm FFL**, accessible, never inside toilet, kitchen wet zone, wardrobe back, or behind door swing.
2. Duplex/floor-wise: one Sub-DB per floor near staircase landing/lobby, fed from Main DB.
3. Commercial: DB per zone/tenancy + emergency lighting circuit separate; DB room/electrical shaft per NBC.
4. Clear access 750mm in front; circuit directory label pasted inside DB door.

Room-wise circuit schedule (mandatory table):

Ckt ID	Description	Rooms/Points covered	Points	Load (W)	MCB	Wire (sq mm)
L1	Lights+fans	Living + foyer	8	700	6A	1.5
L2	Lights+fans	MBR + toilet	...	...	6A	1.5
P1	16A power	Living TV + sockets	2	2000	16A	2.5
AC1	AC MBR	dedicated	1	1800	20A	4.0
GYS1	Geyser T1	dedicated	1	2000	20A	4.0
K1	Kitchen power	counter ring	...	...	16A	2.5/4.0

- Every room's light/fan on a lighting circuit $\leq$ 800W/10 points; adjacent rooms may share, but **toilet exhaust+light stays with its bedroom circuit** for logical tripping.
- Kitchen: min 2 power circuits (counter + fridge dedicated). Every AC and geyser = dedicated circuit.
- Number and mark circuits on the layout drawing itself (point $\rightarrow$ circuit tag).

PART 4 — RCP (REFLECTED CEILING PLAN) + LEGEND + NUMBERING (MANDATORY)

4.1 Standard Legend (use these symbols/codes on RCP and all schedules)

Code	Symbol (drawing)	Description
SL	⊙ (small circle, crosshair)	COB spotlight, recessed
ASL	⊙→	Adjustable/tilt spotlight
PL	■ (square with X)	LED panel light, recessed
CSL	⊗ (double circle)	Ceiling surface light
CV	══ (dashed double line)	Cove LED strip, indirect
WL	● (half-filled circle on wall)	Wall light / sconce
BWL	●b	Bedside wall light
HL	⬡ (hexagon + drop line)	Hanging / pendant light
CH	★ (star in circle)	Chandelier
ML	□m	Mirror/vanity light
STL	•s	Step/foot light
TR	—●—	Magnetic/profile track
CF	× in circle (fan symbol)	Ceiling fan with regulator
EF	⊗	Exhaust fan (wall/ceiling)
AC	□ AC	AC indoor unit (HW=high wall, CST=cassette)
SB	□ + height tag	Switchboard (module count noted)
DB	■ DB	Distribution board

4.2 RCP Rules

1. Produce RCP room-wise: ceiling design + all fixtures with **unique numbering** — format <RoomCode>-<TypeCode>-<No.> → e.g. LIV-SL-01...08, LIV-CF-01, MBR-CV-01, DIN-CH-01.
2. Room codes: LIV, DIN, MBR, BR2/BR3, KIT, TL1/TL2, PJA, STD, FOY, PAS, BAL, UTL (extend for commercial: OFF, CON, REC...).
3. Every fixture number carries: type, W, CCT, mounting height/ceiling drop, dimension from nearest 2 walls (setting-out).

- False ceiling levels marked (+ve from FFL), cove direction arrows, AC linear diffuser/slots coordinated — no clash between fan, chandelier, AC throw, and spot rows.

4.3 Switchboard Schedule mapped to Light Numbering (MANDATORY table)

SB No.	Location + Height	Switch/Module	Controls (fixture nos.)	Circuit
LIV-SB-01	Entry, lock side @1200mm	S1	LIV-PL-01 to 04	L1
		S2	LIV-CV-01	L1
		S3 (dimmer)	LIV-CH-01	L1
		S4 + Reg	LIV-CF-01	L1
LIV-SB-02	Sofa side @300mm	6A ×2	—	P1

Every switch must map to specific fixture numbers — no orphan switches, no uncontrolled fixtures. 2-way pairs shown linked.

PART 5 — AIR CONDITIONING DESIGN + TONNAGE CALCULATION (MEP)

5.1 Room-wise Heat Load / Tonnage (standard format — produce this table)

Room	L×W (ft)	Area (sq ft)	Ht (ft)	Exposure/Glass	Occupancy	Base TR	Corrections	Final TR	Unit Selected
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Method (quick-estimate, flag if detailed HAP/E20 needed):

- Base: $\text{Area (sq ft)} \div 600 = \text{TR}$ (residential, 10ft ceiling) — or 400–500 sq ft/TR for heavy glass/west exposure/commercial.
- Corrections: +0.5 TR west/south wall or large glazing; +0.1 TR per person beyond 2; +5% per ft ceiling height above 10ft; kitchen-open living +0.5 TR; top floor (roof exposure) +10–20%.
- Round to available sizes: 0.8 / 1.0 / 1.5 / 2.0 / 2.2 TR (splits) or VRV indoor sizes.
- Commercial: use occupancy/glass-based check (office ~100 sq ft/person) and recommend detailed heat-load run (HAP/E20) before procurement.

5.2 System Selection — VRV/VRF vs Individual Units (provide BOTH options)

Option 1 — VRV/VRF:

- Sum of indoor TR × **diversity 0.7–0.85** (residential 0.7–0.75, office 0.8–0.9) = outdoor HP/TR selection.
- Show: indoor unit schedule (high-wall/cassette/ductable per room), refnet piping concept, outdoor unit location (terrace/service ledge with ventilation clearance,

piping length/level difference within manufacturer limits), drain routing slope 1:100.

- Best when: total $\geq$ 8-10 TR, ceiling space available, single-owner operation, luxe interiors (concealed/grille-only look).

Option 2 — Individual split/window units:

- One unit per room per 5.1 table, no diversity (each sized standalone).
- Recommend 5-star inverter splits; window AC only for budget/retrofit cases.

5.3 Indoor Unit Location Rules (per furniture orientation)

1. **High-wall split:** long throw along room length; **bedroom — never blowing directly on bed/pillow**; mount on wall perpendicular to bed head or throwing across foot side; @ 2100-2250mm to unit bottom, min 150mm below ceiling.
2. **Living:** throw across seating zone, not point-blank at sofa backs; no unit above TV panel; coordinate with curtain pelmet.
3. **Cassette:** seating-zone center on RCP, min 1.2m clear from fan/chandelier.
4. **Outdoor units:** service ledge/terrace, clearances 300mm rear / 1000mm front, drain to nearest trap, not facing neighbor openings at close range.
5. **Every unit:** dedicated power point + isolator at unit level, drain line + copper piping route + wall/beam sleeve marked on layout.

5.4 AC SCHEDULE (mandatory output table)

AC No.	Room	Type (HW/CST/Ductable/Window)	TR	kW	Power Point Ht	Circuit ID	Drain Route	Remarks
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5.5 DB SCHEDULE — FINAL CONSOLIDATED (mandatory output table)

DB ID	Ckt	Description	Phase	Points	Connected W	Demand W	MCB	RCCB Bank	Wire sq mm	Remarks
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Close with: total connected load, demand load, phase balance summary, recommended sanctioned load, and main switch/MCCB + incoming cable size from meter to Main DB.

PART 6 — MOOD LIGHTING OPTIONS (provide as alternates for same layout)

Option A — CLASSIC WARM LUXE: All 2700-3000K, cove-dominant ambient, spots dimmable, statement chandeliers/pendants, wall sconces. 3-scene: Full / Evening (cove+accent only) / Night (foot lights + 10%).

Option B — MODERN MINIMAL: Magnetic track + linear profile language, 3500K, minimal cove, no decorative pendants, hidden light sources only. Scenes: Work / Relax / Away.

Option C — SMART TUNABLE: Tunable white 2700–5700K + RGB accent coves, app/voice control, circadian schedule (cool morning → warm night), motion-sensor corridor/toilet, all key circuits on dimmer modules.

Option D — HOSPITALITY DRAMA (commercial/luxe): High contrast — ambient 50–100 lux, accents 3–5× on art/textures, grazing on stone walls, 2400–2700K, DALI/0–10V dimming zones.

For each option give: revised fixture schedule, control gear (dimmer type), added load delta, and budget impact (Low/Med/High).

DELIVERABLES CHECKLIST (AI must output all)

1. Room-wise lighting design narrative + lux verification calc
2. Fixture schedule (type, W, CCT, beam, qty, mounting)
3. Ceiling layout — fan positions (qty per room rule 1.3), chandelier/pendant positions, light placement in standard peripheral format
4. **RCP with legend, symbols, and fixture numbering** (LIV-SL-01 format) + setting-out dimensions
5. **Switchboard schedule mapped to light numbering** — every switch → fixture nos.
6. Switchboard/point schedule — location, height, modules, rating, room-wise
7. Load calculation + diversity + circuit division
8. **SLD** — meter to final circuits with ratings, wire sizes, phase balancing
9. **DB location plan + consolidated DB schedule** with circuit IDs tagged to layout
10. **AC tonnage calculation table (room-wise)** + VRV vs individual unit options with diversity + indoor/outdoor unit locations per furniture orientation
11. **AC schedule** (unit, TR, power point, circuit, drain)
12. Mood lighting options A–D comparison table
13. Compliance notes: NBC 2016, IS 732, IS 3646, IS 3043 earthing, ECBC LPD, RCD protection, IP ratings in wet areas
14. Assumptions list + items needing client confirmation